

# SUNIX SDC Expansion Board

## SDC Linux Library

## Development Manual

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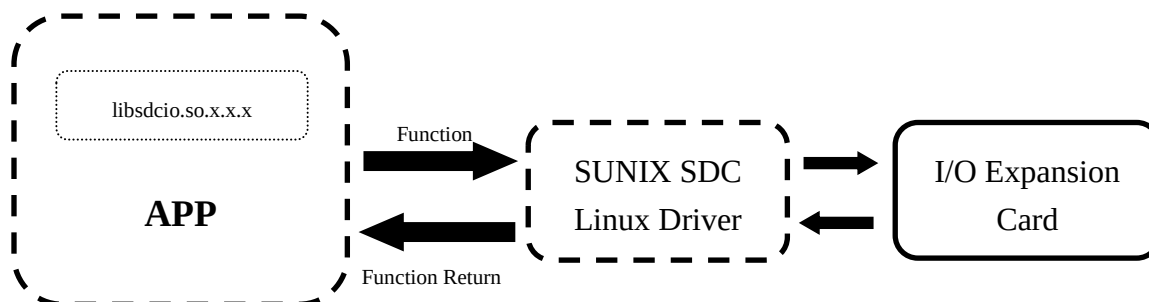
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# 1. Introduction

The SDC library can help users develop program to control SUNIX IO Expansion Card more easily.



## 1.1. Requirement

Support to Linux kernel 4.x/5.x/6.x x86-64 or arm64 platform.

## 1.2. Library Installation

Login as root, copy **libsdcio.so.x.x.x** to **/usr/lib**.

The .so file position is:

**./SDCLinuxExpansionBoardDriverVX.X.X.X/lib/files/x86-64** or

**./SDCLinuxExpansionBoardDriverVX.X.X.X/lib/files/arm64**

```
$cp libsdci.so.x.x.x /usr/lib
```

## 1.3. Before Development For Your C Project

- 1). Copy **sdclib.h** to C project directory and include it.
- 2). Enter command:

```
$ldconfig
```

# 2. SDC Functions

## 2.1. SDC Function List

| DIO Function                          | Description                                                |
|---------------------------------------|------------------------------------------------------------|
| 1). <b>sdc_dio_open</b>               | DIO channel initial.                                       |
| 2). <b>sdc_dio_close</b>              | Free DIO channel.                                          |
| 3). <b>sdc_dio_get_info</b>           | Get DIO channel information, e.g. line number, bank index. |
| 4). <b>sdc_dio_get_bank_direction</b> | Get DIO direction, input or output.                        |

|                                                             |                                                                 |
|-------------------------------------------------------------|-----------------------------------------------------------------|
| 5). sdc_dio_set_bank_direction                              | Set DIO direction, input or output.                             |
| 6). sdc_dio_get_bank_state                                  | Get DIO state, high or low.                                     |
| 7). sdc_dio_set_bank_state                                  | Set DIO state, high or low.                                     |
| 8). sdc_dio_get_bank_input_invert                           | Get DI bank invert, disable or enable.                          |
| 9). sdc_dio_set_bank_input_invert                           | Set DI bank invert, disable or enable.                          |
| 10). sdc_dio_get_bank_input_latch_positive_edge             | Get DI bank positive edge latch, happened or never happened.    |
| 11). sdc_dio_set_bank_input_latch_positive_edge             | Reset DI bank positive edge latch, reset or never do not reset. |
| 12). sdc_dio_get_bank_input_latch_negative_edge             | Get DI bank negative edge latch, happened or never happened.    |
| 13). sdc_dio_set_bank_input_latch_negative_edge             | Reset DI bank negative edge latch, reset or do not reset.       |
| 14). sdc_dio_set_bank_input_counter_reset                   | DI bank counter reset.                                          |
| 15). sdc_dio_get_bank_input_counter_increment_positive_edge | Get DI bank positive edge counter, disable or enable.           |
| 16). sdc_dio_set_bank_input_counter_increment_positive_edge | Set DI bank positive edge counter, disable or enable.           |
| 17). sdc_dio_get_bank_input_counter_increment_negative_edge | Get DI bank negative edge counter, disable or enable.           |
| 18). sdc_dio_set_bank_input_counter_increment_negative_edge | Set DI bank negative edge counter, disable or enable.           |
| 19). sdc_dio_get_bank_input_port_counter_value              | Get DI bank single DO port counter value.                       |
| 20). sdc_dio_get_bank_input_event_ctrl                      | Get DI bank input rising and falling event control.             |
| 21). sdc_dio_set_bank_input_event_ctrl                      | Set DI bank input rising and falling event control.             |
| 22). sdc_dio_get_bank_input_filter_value                    | Get DI bank filter timer. Unit: millisecond.                    |
| 23). sdc_dio_set_bank_input_filter_value                    | Set DI bank filter timer. Unit: millisecond.                    |
| 24). sdc_dio_register_event_callback                        | Register DI bank event callback function.                       |
| 25). Sdc_dio_unregister_event_callback                      | Unregister DI bank event callback function.                     |
| 26). sdc_dio_get_bank_output_initial_value                  | Get DO bank initial value.                                      |
| 27). sdc_dio_set_bank_output_initial_value                  | Set DO bank initial value.                                      |

## 2.2. SDC DIO Functions

### 2.2.1. sdc\_dio\_open

- Syntax:

```
int sdc_dio_open(int line);
```

- Paramter:

```
line[in]                      Line number, Device node “sdc_dion”, n = line number.
```

- Return Code:

If the function is successful, the return value is zero.  
If the function fails, the return value is error code. Please check the return codes table.

### 2.2.2. sdc\_dio\_close

- Syntax:

```
int sdc_dio_close(int line);
```

- Paramter:

|          |                                                                 |
|----------|-----------------------------------------------------------------|
| line[in] | Line number, Device node “sdc_dion”, $n = \text{line number}$ . |
|----------|-----------------------------------------------------------------|

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

### 2.2.3. sdc\_dio\_get\_info

- Syntax:

```
int sdc_dio_get_info(
_in_ int line,
_out_ struct sdc_dio_info *info
);
```

- Paramter:

|           |                                                                 |
|-----------|-----------------------------------------------------------------|
| line[in]  | Line number, Device node “sdc_dion”, $n = \text{line number}$ . |
| info[out] | DIO channel information structure pointer.                      |

➤ Refer to chapter 3.1 **\_SDC\_DIO\_INFO** and 3.2 **\_SDC\_BANK\_INFO**

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

### 2.2.4. sdc\_dio\_get\_bank\_direction

- Syntax:

```
int sdc_dio_get_bank_direction(
_in_ int line,
_in_ int bank_index,
_out_ int *direction
);
```

- Paramter:

|                |                                                                                                                                                             |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| line[in]       | Line number, Device node “sdc_dion”, $n = \text{line number}$ .                                                                                             |
| bank_index[in] | Bank index, the layer determine whether is DI or DO.<br>bank_index = 0, cap_input = 1, the bank0 is DI.<br>bank_index = 1, cap_output = 1, the bank1 is DO. |
| direction[out] | DIO direction, 0 is input, 1 is output.                                                                                                                     |

e.g. 8DIO device, direction = 000000FF  
= 0000 0000 0000 0000 0000 0000 1111 1111 (binary)  
→ bit0~7: DIO1~8 are output

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

### 2.2.5. sdc\_dio\_set\_bank\_direction

- Syntax:

```
int sdc_dio_set_bank_direction(
_in_ int line,
_in_ int bank_index,
_in_ int direction
);
```

- Paramter:

|                |                                                                                                                                                                        |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| line[in]       | Line number, Device node “sdc_dion”, $n = \text{line number}$ .                                                                                                        |
| bank_index[in] | Bank index, the layer determine whether is DI or DO.<br>bank_index = 0, cap_input = 1, the bank0 is DI.<br>bank_index = 1, cap_output = 1, the bank1 is DO.            |
| direction[in]  | DIO direction, 0 is input, 1 is output.<br>e.g. 8DIO device, direction = 000000FF<br>= 0000 0000 0000 0000 0000 0000 1111 1111 (binary)<br>→ bit0~7: DIO1~8 are output |

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

### 2.2.6. sdc\_dio\_get\_bank\_state

- Syntax:

```
int sdc_dio_get_bank_state(
_in_ int line,
_in_ int bank_index,
_out_ int *state
);
```

- Paramter:

|                |                                                                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| line[in]       | Line number, Device node “sdc_dion”, $n = \text{line number}$ .                                                                                                                                                                           |
| bank_index[in] | Bank index, the layer determine whether is DI or DO.<br>bank_index = 0, cap_input = 1, the bank0 is DI.<br>bank_index = 1, cap_output = 1, the bank1 is DO.                                                                               |
| state[out]     | DIO state, 0 is low, 1 is high.<br>e.g. 8DI device, state = 0x000000AA<br>= 0000 0000 0000 0000 0000 0000 1010 1010 (binary)<br>→ bit0, bit2, bit4, bit6: DI1, 3, 5, and 7 are low<br>→ bit1, bit3, bit5, bit7: DI2, 4, 6, and 8 are high |

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

#### 2.2.7. sdc\_dio\_set\_bank\_state

- Syntax:

```
int sdc_dio_set_bank_state(
_in_ int line,
_in_ int bank_index,
_in_ int state
);
```

- Paramter:

|                |                                                                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| line[in]       | Line number, Device node “sdc_dion”, $n = \text{line number}$ .                                                                                                                                                                           |
| bank_index[in] | Bank index, the layer determine whether is DI or DO.<br>bank_index = 0, cap_input = 1, the bank0 is DI.<br>bank_index = 1, cap_output = 1, the bank1 is DO.                                                                               |
| state[in]      | DIO state, 0 is low, 1 is high.<br>e.g. 8DO device, state = 0x000000AA<br>= 0000 0000 0000 0000 0000 0000 1010 1010 (binary)<br>→ bit0, bit2, bit4, bit6: DO1, 3, 5, and 7 are low<br>→ bit1, bit3, bit5, bit7: DO2, 4, 6, and 8 are high |

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

#### 2.2.8. sdc\_dio\_get\_bank\_input\_invert

- Syntax:



```
int sdc_dio_get_bank_input_invert(
_in_ int line,
_in_ int bank_index,
_out_ int *invert
);
```

- Paramter:

|                |                                                                                                                                                                                                                                                                   |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| line[in]       | Line number, Device node “sdc_dion”, $n = \text{line number}$ .                                                                                                                                                                                                   |
| bank_index[in] | Bank index, the layer determine whether is DI or DO.<br>bank_index = 0, cap_input = 1, the bank0 is DI.<br>bank_index = 1, cap_output = 1, the bank1 is DO.                                                                                                       |
| invert[out]    | DI bank input invert, 0 is disable, 1 is enable.<br>e.g. 8DI device, invert = 0x000000AA<br>= 0000 0000 0000 0000 0000 0000 1010 1010 (binary)<br>→ bit0, bit2, bit4, bit6: DI1, 3, 5, and 7 are disable<br>→ bit1, bit3, bit5, bit7: DI2, 4, 6, and 8 are enable |

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

### 2.2.9. sdc\_dio\_set\_bank\_input\_invert

- Syntax:

```
int sdc_dio_set_bank_input_invert(
_in_ int line,
_in_ int bank_index,
_in_ int invert
);
```

- Paramter:

|                |                                                                                                                                                                                                                                                                   |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| line[in]       | Line number, Device node “sdc_dion”, $n = \text{line number}$ .                                                                                                                                                                                                   |
| bank_index[in] | Bank index, the layer determine whether is DI or DO.<br>bank_index = 0, cap_input = 1, the bank0 is DI.<br>bank_index = 1, cap_output = 1, the bank1 is DO.                                                                                                       |
| invert[in]     | DI bank input invert, 0 is disable, 1 is enable.<br>e.g. 8DI device, invert = 0x000000AA<br>= 0000 0000 0000 0000 0000 0000 1010 1010 (binary)<br>→ bit0, bit2, bit4, bit6: DI1, 3, 5, and 7 are disable<br>→ bit1, bit3, bit5, bit7: DI2, 4, 6, and 8 are enable |

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

#### 2.2.10. sdc\_dio\_get\_bank\_input\_latch\_positive\_edge

- Syntax:

```
int sdc_dio_get_bank_input_latch_positive_edge(
_in_ int line,
_in_ int bank_index,
_out_ int *latch
);
```

- Paramter:

|                |                                                                                                                                                                                                                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| line[in]       | Line number, Device node “sdc_dion”, $n = \text{line number}$ .                                                                                                                                                                                                                                 |
| bank_index[in] | Bank index, the layer determine whether is DI or DO.<br>bank_index = 0, cap_input = 1, the bank0 is DI.<br>bank_index = 1, cap_output = 1, the bank1 is DO.                                                                                                                                     |
| latch[in]      | DI bank input positive edge latch, 0 is never happened, 1 is happened.<br>e.g. 8DI device, latch = 0x000000AA<br>= 0000 0000 0000 0000 0000 0000 1010 1010 (binary)<br>→ bit0, bit2, bit4, bit6: DI1, 3, 5, and 7 are never happened<br>→ bit1, bit3, bit5, bit7: DI2, 4, 6, and 8 are happened |

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

#### 2.2.11. sdc\_dio\_set\_bank\_input\_latch\_positive\_edge

- Syntax:

```
int sdc_dio_set_bank_input_latch_positive_edge(
_in_ int line,
_in_ int bank_index,
_in_ int latch_reset
);
```

- Paramter:

|                |                                                                 |
|----------------|-----------------------------------------------------------------|
| line[in]       | Line number, Device node “sdc_dion”, $n = \text{line number}$ . |
| bank_index[in] | Bank index, the layer determine whether is DI or DO.            |

|                  |                                                                                                                                                                                                                                                                                  |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | bank_index = 0, cap_input = 1, the bank0 is DI.<br>bank_index = 1, cap_output = 1, the bank1 is DO.                                                                                                                                                                              |
| latch_reset [in] | DI bank input positive latch reset, 0 is do not reset, 1 is reset.<br>e.g. 8DI device, reset = 0x000000AA<br>= 0000 0000 0000 0000 0000 0000 1010 1010 (binary)<br>→ bit0, bit2, bit4, bit6: DI1, 3, 5, and 7 does not reset<br>→ bit1, bit3, bit5, bit7: DI2, 4, 6, and 8 reset |

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

### 2.2.12. sdc\_dio\_get\_bank\_input\_latch\_negative\_edge

- Syntax:

```
int sdc_dio_get_bank_input_latch_negative_edge(
_in_ int line,
_in_ int bank_index,
_out_ int *latch
);
```

- Paramter:

|                |                                                                                                                                                                                                                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| line[in]       | Line number, Device node “sdc_dion”, $n = \text{line number}$ .                                                                                                                                                                                                                                 |
| bank_index[in] | Bank index, the layer determine whether is DI or DO.<br>bank_index = 0, cap_input = 1, the bank0 is DI.<br>bank_index = 1, cap_output = 1, the bank1 is DO.                                                                                                                                     |
| latch[in]      | DI bank input negative edge latch, 0 is never happened, 1 is happened.<br>e.g. 8DI device, latch = 0x000000AA<br>= 0000 0000 0000 0000 0000 0000 1010 1010 (binary)<br>→ bit0, bit2, bit4, bit6: DI1, 3, 5, and 7 are never happened<br>→ bit1, bit3, bit5, bit7: DI2, 4, 6, and 8 are happened |

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

### 2.2.13. sdc\_dio\_set\_bank\_input\_latch\_negative\_edge

- Syntax:

```
int sdc_dio_set_bank_input_latch_negative_edge(
_in_ int line,
```

```
_in_ int bank_index,  
_in_ int latch_reset  
);
```

- Paramter:

|                  |                                                                                                                                                                                                                                                                                       |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| line[in]         | Line number, Device node “sdc_dion”, $n = \text{line number}$ .                                                                                                                                                                                                                       |
| bank_index[in]   | Bank index, the layer determine whether is DI or DO.<br>bank_index = 0, cap_input = 1, the bank0 is DI.<br>bank_index = 1, cap_output = 1, the bank1 is DO.                                                                                                                           |
| latch_reset [in] | DI bank input negative edge latch reset, 0 is do not reset, 1 is reset.<br>e.g. 8DI device, reset = 0x000000AA<br>= 0000 0000 0000 0000 0000 0000 1010 1010 (binary)<br>→ bit0, bit2, bit4, bit6: DI1, 3, 5, and 7 does not reset<br>→ bit1, bit3, bit5, bit7: DI2, 4, 6, and 8 reset |

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

#### 2.2.14. sdc\_dio\_set\_bank\_input\_counter\_reset

- Syntax:

```
int sdc_dio_set_bank_input_counter_reset(  
_in_ int line,  
_in_ int bank_index,  
_in_ int counter_reset  
);
```

- Paramter:

|                    |                                                                                                                                                                                                                                                                           |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| line[in]           | Line number, Device node “sdc_dion”, $n = \text{line number}$ .                                                                                                                                                                                                           |
| bank_index[in]     | Bank index, the layer determine whether is DI or DO.<br>bank_index = 0, cap_input = 1, the bank0 is DI.<br>bank_index = 1, cap_output = 1, the bank1 is DO.                                                                                                               |
| counter_reset [in] | DI bank input counter reset, 0 is do not reset, 1 is reset.<br>e.g. 8DI device, reset = 0x000000AA<br>= 0000 0000 0000 0000 0000 0000 1010 1010 (binary)<br>→ bit0, bit2, bit4, bit6: DI1, 3, 5, and 7 does not reset<br>→ bit1, bit3, bit5, bit7: DI2, 4, 6, and 8 reset |

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

#### 2.2.15. sdc\_dio\_get\_bank\_input\_counter\_increment\_positive\_edge

- Syntax:

```
int sdc_dio_get_bank_input_counter_increment_positive_edge(
_in_ int line,
_in_ int bank_index,
_out_ int *counter
);
```

- Paramter:

|                |                                                                                                                                                                                                                                                                                        |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| line[in]       | Line number, Device node “sdc_dion”, $n = \text{line number}$ .                                                                                                                                                                                                                        |
| bank_index[in] | Bank index, the layer determine whether is DI or DO.<br>bank_index = 0, cap_input = 1, the bank0 is DI.<br>bank_index = 1, cap_output = 1, the bank1 is DO.                                                                                                                            |
| counter[out]   | DI bank input positive edge counter enable, 0 is disable, 1 is enable.<br>e.g. 8DI device, latch = 0x000000AA<br>= 0000 0000 0000 0000 0000 0000 1010 1010 (binary)<br>→ bit0, bit2, bit4, bit6: DI1, 3, 5, and 7 are disable<br>→ bit1, bit3, bit5, bit7: DI2, 4, 6, and 8 are enable |

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

#### 2.2.16. sdc\_dio\_set\_bank\_input\_counter\_increment\_positive\_edge

- Syntax:

```
int sdc_dio_set_bank_input_counter_increment_positive_edge(
_in_ int line,
_in_ int bank_index,
_in_ int counter
);
```

- Paramter:

|                |                                                                                                                                                             |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| line[in]       | Line number, Device node “sdc_dion”, $n = \text{line number}$ .                                                                                             |
| bank_index[in] | Bank index, the layer determine whether is DI or DO.<br>bank_index = 0, cap_input = 1, the bank0 is DI.<br>bank_index = 1, cap_output = 1, the bank1 is DO. |

|             |                                                                                                                                                                                                                                                                                        |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| counter[in] | DI bank input positive edge counter enable, 0 is disable, 1 is enable.<br>e.g. 8DI device, latch = 0x000000AA<br>= 0000 0000 0000 0000 0000 0000 1010 1010 (binary)<br>→ bit0, bit2, bit4, bit6: DI1, 3, 5, and 7 are disable<br>→ bit1, bit3, bit5, bit7: DI2, 4, 6, and 8 are enable |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

### 2.2.17. sdc\_dio\_get\_bank\_input\_counter\_increment\_negative\_edge

- Syntax:

```
int sdc_dio_get_bank_input_counter_increment_negative_edge(
    _in_ int line,
    _in_ int bank_index,
    _out_ int *counter
);
```

- Paramter:

|                |                                                                                                                                                                                                                                                                                        |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| line[in]       | Line number, Device node “sdc_dion”, $n = \text{line number}$ .                                                                                                                                                                                                                        |
| bank_index[in] | Bank index, the layer determine whether is DI or DO.<br>bank_index = 0, cap_input = 1, the bank0 is DI.<br>bank_index = 1, cap_output = 1, the bank1 is DO.                                                                                                                            |
| counter[out]   | DI bank input negative edge counter enable, 0 is disable, 1 is enable.<br>e.g. 8DI device, latch = 0x000000AA<br>= 0000 0000 0000 0000 0000 0000 1010 1010 (binary)<br>→ bit0, bit2, bit4, bit6: DI1, 3, 5, and 7 are disable<br>→ bit1, bit3, bit5, bit7: DI2, 4, 6, and 8 are enable |

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

### 2.2.18. sdc\_dio\_set\_bank\_input\_counter\_increment\_negative\_edge

- Syntax:

```
int sdc_dio_set_bank_input_counter_increment_negative_edge(
    _in_ int line,
    _in_ int bank_index,
    _in_ int counter
);
```

```
);
```

- Paramter:

|                |                                                                                                                                                                                                                                                                                        |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| line[in]       | Line number, Device node “sdc_dion”, $n = \text{line number}$ .                                                                                                                                                                                                                        |
| bank_index[in] | Bank index, the layer determine whether is DI or DO.<br>bank_index = 0, cap_input = 1, the bank0 is DI.<br>bank_index = 1, cap_output = 1, the bank1 is DO.                                                                                                                            |
| counter[in]    | DI bank input negative edge counter enable, 0 is disable, 1 is enable.<br>e.g. 8DI device, latch = 0x000000AA<br>= 0000 0000 0000 0000 0000 0000 1010 1010 (binary)<br>→ bit0, bit2, bit4, bit6: DI1, 3, 5, and 7 are disable<br>→ bit1, bit3, bit5, bit7: DI2, 4, 6, and 8 are enable |

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

#### 2.2.19. sdc\_dio\_get\_bank\_input\_counter\_value

- Syntax:

```
int sdc_dio_get_bank_input_counter_value(
_in_ int line,
_in_ int bank_index,
_in_ int port_index,
_out_ int *counter_value
);
```

- Paramter:

|                     |                                                                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| line[in]            | Line number, Device node “sdc_dion”, $n = \text{line number}$ .                                                                                             |
| bank_index[in]      | Bank index, the layer determine whether is DI or DO.<br>bank_index = 0, cap_input = 1, the bank0 is DI.<br>bank_index = 1, cap_output = 1, the bank1 is DO. |
| port_index[in]      | DI port index, the range is 0x00~0x1F(0~31).                                                                                                                |
| counter_value [out] | DI single port counter value = positive counter + negative counter.                                                                                         |

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

#### 2.2.20. sdc\_dio\_get\_bank\_input\_event\_ctrl

- Syntax:

```
int sdc_dio_get_bank_input_event_ctrl(
_in_ int line,
_in_ int bank_index,
_out_ int *rising,
_out_ int *falling
);
```

- Paramter:

|                |                                                                                                                                                                                                                                                                                     |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| line[in]       | Line number, Device node “sdc_dion”, $n$ = line number.                                                                                                                                                                                                                             |
| bank_index[in] | Bank index, the layer determine whether is DI or DO.<br>bank_index = 0, cap_input = 1, the bank0 is DI.<br>bank_index = 1, cap_output = 1, the bank1 is DO.                                                                                                                         |
| rising[out]    | DI bank input rising event mode enable, 0 is disable, 1 is enable.<br>e.g. 8DI device, latch = 0x000000AA<br>= 0000 0000 0000 0000 0000 0000 1010 1010 (binary)<br>→ bit0, bit2, bit4, bit6: DI1, 3, 5, and 7 are disable<br>→ bit1, bit3, bit5, bit7: DI2, 4, 6, and 8 are enable  |
| falling[out]   | DI bank input falling event mode enable, 0 is disable, 1 is enable.<br>e.g. 8DI device, latch = 0x000000AA<br>= 0000 0000 0000 0000 0000 0000 1010 1010 (binary)<br>→ bit0, bit2, bit4, bit6: DI1, 3, 5, and 7 are disable<br>→ bit1, bit3, bit5, bit7: DI2, 4, 6, and 8 are enable |

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

### 2.2.21. sdc\_dio\_set\_bank\_input\_event\_ctrl

- Syntax:

```
int sdc_dio_set_bank_input_event_ctrl(
_in_ int line,
_in_ int bank_index,
_in_ int rising,
_in_ int falling
);
```

- Paramter:

|          |                                                         |
|----------|---------------------------------------------------------|
| line[in] | Line number, Device node “sdc_dion”, $n$ = line number. |
|----------|---------------------------------------------------------|



|                |                                                                                                                                                                                                                                                                                 |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bank_index[in] | Bank index, the layer determine whether is DI or DO.<br>bank_index = 0, cap_input = 1, the bank0 is DI.<br>bank_index = 1, cap_output = 1, the bank1 is DO.                                                                                                                     |
| rising [in]    | DI bank input rising event control, 0 is disable, 1 is enable.<br>e.g. 8DI device, latch = 0x000000AA<br>= 0000 0000 0000 0000 0000 0000 1010 1010 (binary)<br>→ bit0, bit2, bit4, bit6: DI1, 3, 5, and 7 are disable<br>→ bit1, bit3, bit5, bit7: DI2, 4, 6, and 8 are enable  |
| falling[in]    | DI bank input falling event control, 0 is disable, 1 is enable.<br>e.g. 8DI device, latch = 0x000000AA<br>= 0000 0000 0000 0000 0000 0000 1010 1010 (binary)<br>→ bit0, bit2, bit4, bit6: DI1, 3, 5, and 7 are disable<br>→ bit1, bit3, bit5, bit7: DI2, 4, 6, and 8 are enable |

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

## 2.2.22. sdc\_dio\_get\_bank\_input\_filter\_value

- Syntax:

```
int sdc_dio_get_bank_input_filter_value (
_in_ int line,
_in_ int bank_index,
_in_ int port_index,
_out_ int *filter_timer
);
```

- Paramter:

|                   |                                                                                                                                                             |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| line[in]          | Line number, Device node “sdc_dion”, $n = \text{line number}$ .                                                                                             |
| bank_index[in]    | Bank index, the layer determine whether is DI or DO.<br>bank_index = 0, cap_input = 1, the bank0 is DI.<br>bank_index = 1, cap_output = 1, the bank1 is DO. |
| port_index[in]    | DI port index, the range is 0x00~0x1F(0~31).                                                                                                                |
| filter_timer[out] | DI filter timer, Unit:ms, min = 1ms, max = 10s = 10000ms<br>e.g. if timer = 1, filter value = 1ms<br>if timer = 10000, filter value = 10s                   |

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the error codes table.

### 2.2.23. sdc\_dio\_set\_bank\_input\_filter\_value

- Syntax:

```
int sdc_dio_set_bank_input_filter_value (
    _in_ int line,
    _in_ int bank_index,
    _in_ int port_index,
    _in_ int filter_timer
);
```

- Paramter:

|                  |                                                                                                                                                             |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| line[in]         | Line number, Device node “sdc_dion”, $n = \text{line number}$ .                                                                                             |
| bank_index[in]   | Bank index, the layer determine whether is DI or DO.<br>bank_index = 0, cap_input = 1, the bank0 is DI.<br>bank_index = 1, cap_output = 1, the bank1 is DO. |
| port_index[in]   | DI port index, the range is 0x00~0x1F(0~31).                                                                                                                |
| filter_timer[in] | DI filter timer, Unit:ms, min = 1ms, max = 10s = 10000ms<br>e.g. if timer = 1, filter value = 1ms<br>if timer = 10000, filter value = 10s                   |

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

### 2.2.24. sdc\_dio\_register\_event\_callback

- Syntax:

```
void sdc_dio_register_event_callback(
    _in_ int line,
    _in_ sdc_dio_callback callback_ptr
);
type void (*sdc_dio_callback)( int line, int bank_index, int input_value, int input_delta)
```

- Paramter:

|                  |                                                                 |                                                                                                         |
|------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| line[in]         | Line number, Device node “sdc_dion”, $n = \text{line number}$ . |                                                                                                         |
| callback_ptr[in] | SDC DIO callback function pointer.                              |                                                                                                         |
|                  | line                                                            | Line number, Device node “sdc_dion”, $n = \text{line number}$ .                                         |
|                  | bank_index                                                      | Bank index, the layer determine whether is DI or DO.<br>bank_index = 0, cap_input = 1, the bank0 is DI. |

|  |             |                                                                                                                                                                                                                                        |
|--|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |             | bank_index = 1, cap_output = 1, the bank1 is DO.                                                                                                                                                                                       |
|  | input_value | DI value after state change.<br>e.g. 8DI device, state = 0x000000AA<br>= 0000 0000 0000 0000 0000 0000 1010 1010 (binary)<br>→ bit0, bit2, bit4, bit6: DI1, 3, 5, and 7 are low<br>→ bit1, bit3, bit5, bit7: DI2, 4, 6, and 8 are high |
|  | input_delta | DI delta.<br>e.g. 8DI device, delta = 0x000000AA<br>= 0000 0000 0000 0000 0000 0000 1010 1010 (binary)<br>→ bit0, bit2, bit4, bit6: DIO1, 3, 5, and 7 are unchanged<br>→ bit1, bit3, bit5, bit7: DIO2, 4, 6, and 8 are changed         |

#### 2.2.25. sdc\_dio\_unregister\_event\_callback

- Syntax:

```
void sdc_dio_unregister_event_callback(
_in_ int line
);
```

- Paramter:

|          |                                                                 |
|----------|-----------------------------------------------------------------|
| line[in] | Line number, Device node “sdc_dion”, $n = \text{line number}$ . |
|----------|-----------------------------------------------------------------|

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

#### 2.2.26. sdc\_dio\_get\_output\_initial\_value

- Syntax:

```
void sdc_dio_get_output_initial_value (
_in_ int line,
_in_ int bank_index,
_out_ int *init_value
);
```

- Paramter:

|                 |                                                                                                                                                             |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| line[in]        | Line number, Device node “sdc_dion”, $n = \text{line number}$ .                                                                                             |
| bank_index[in]  | Bank index, the layer determine whether is DI or DO.<br>bank_index = 0, cap_input = 1, the bank0 is DI.<br>bank_index = 1, cap_output = 1, the bank1 is DO. |
| init_value[out] | DO initial value, 0 is low, 1 is high.                                                                                                                      |

e.g. 8DO device, state = 0x000000AA  
 = 0000 0000 0000 0000 0000 0000 1010 1010 (binary)  
 → bit0, bit2, bit4, bit6: DI1, 3, 5, and 7 are low  
 → bit1, bit3, bit5, bit7: DI2, 4, 6, and 8 are high

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

### 2.2.27. sdc\_dio\_set\_output\_initial\_value

- Syntax:

```
void sdc_dio_set_output_initial_value (
_in_ int line,
_in_ int bank_index,
_in_ int init_value
);
```

- Paramter:

|                |                                                                                                                                                                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| line[in]       | Line number, Device node “sdc_dion”, $n = \text{line number}$ .                                                                                                                                                                                  |
| bank_index[in] | Bank index, the layer determine whether is DI or DO.<br>bank_index = 0, cap_input = 1, the bank0 is DI.<br>bank_index = 1, cap_output = 1, the bank1 is DO.                                                                                      |
| init_value[in] | DO initial value, 0 is low, 1 is high.<br>e.g. 8DO device, state = 0x000000AA<br>= 0000 0000 0000 0000 0000 0000 1010 1010 (binary)<br>→ bit0, bit2, bit4, bit6: DI1, 3, 5, and 7 are low<br>→ bit1, bit3, bit5, bit7: DI2, 4, 6, and 8 are high |

- Return Code:

If the function is successful, the return value is zero.

If the function fails, the return value is error code. Please check the return codes table.

## 3. Structures

### 3.1. sdc\_dio\_info

- Syntax:

```
struct sdc_dio_info {
    int pci_bus;
    int irq;
```

```

    int line;
    int index;
    int version;
    int cap_samedirection;
    int cap_storeflash;
    int nr_bank;
    int di_sampling_freq;
    int di_filter_lower_bound_ms;
    int di_filter_min_ms;
    int di_filter_max_ms;
    struct sdc_dio_bank_info banks[32]
};

```

- Members:

|                          |                                                            |
|--------------------------|------------------------------------------------------------|
| pci_bus                  | PCI BUS number.                                            |
| irq                      | IRQ number.                                                |
| line                     | Controller line number. “sdc_dion”, $n$ = line number.     |
| index                    | SDC controller index number.                               |
| version                  | Controller version.                                        |
| cap_samedirection        | Bank share same direction capability.                      |
| cap_storeflash           | Store output initial value into FLASH capability.          |
| nr_bank                  | Number of bank in this controller.                         |
| di_sampling_freq         | DI sampling frequency.                                     |
| di_filter_lower_bound_ms | DI sampling lower bound value. Unit: millisecond.          |
| di_filter_min_ms         | DI sampling minimum value. Unit: millisecond.              |
| di_filter_max_ms         | DI sampling maximum value. Unit: millisecond.              |
| banks                    | Array of sdc_dio_bank_info structure. Maximum amount = 32. |

### 3.2. sdc\_dio\_bank\_info

- Syntax:

```

struct sdc_dio_bank_info {
    int nr_port;
    int cap_input;
    int cap_output;
    int cap_rising_trigger;
    int cap_falling_trigger;
}

```

- Members:

|                     |                                              |       |
|---------------------|----------------------------------------------|-------|
| nr_port             | Number of I/O port in this bank.             |       |
| cap_input           | Bank input capability.                       |       |
|                     | Capability                                   | Value |
|                     | It's input                                   | 0     |
|                     | It's not input                               | 1     |
| cap_output          | Bank output capability.                      |       |
|                     | Capability                                   | Value |
|                     | It's output                                  | 0     |
|                     | It's not output                              | 1     |
| cap_rising_trigger  | Bank input event rising trigger capability.  |       |
|                     | Capability                                   | Value |
|                     | Rising trigger disable                       | 0     |
|                     | Rising trigger enable                        | 1     |
| cap_falling_trigger | Bank input event falling trigger capability. |       |
|                     | Capability                                   | Value |
|                     | Falling trigger disable                      | 0     |
|                     | Falling trigger enable                       | 1     |

## 4. Return Code Table

### 4.1. Return Code Table

| Return Code | Describe                            |
|-------------|-------------------------------------|
| 0           | STATUS_SUCCESS                      |
| -1          | STATUS_INVALID_PARAMETER            |
| -2          | STATUS_DEVICE_BUSY                  |
| -3          | STATUS_NO_SUCH_DEVICE               |
| -4          | STATUS_ALLOC_MEMORY_FAIL            |
| -5          | STATUS_CREATE_THREAD_FAIL           |
| -6          | STATUS_IOCTL_FAIL                   |
| -7          | STATUS_CONTROLLER_VERSION_UNSupport |
| -101        | STATUS_DIO_BANK_NOT_FOUND           |
| -102        | STATUS_DIO_BANK_PORT_NOT_FOUND      |
| -103        | STATUS_DIO_BANK_NO_INPUT_CAP        |
| -104        | STATUS_DIO_BANK_NO_OUTPUT_CAP       |